Nova Analytics - Prediction Report

Dataset: auto-mpg.csv

Prediction Date: August 02, 2026 | Prediction Time: 09:04:06 PM

Training Configuration

Selected ML Model	**Polynomial Regression**
Target Column	**cylinders**
Feature Columns	**mpg, displacement**
Train/Test Split	**85% / 15%**
Model Parameters	**degree: 2**

Training Summary

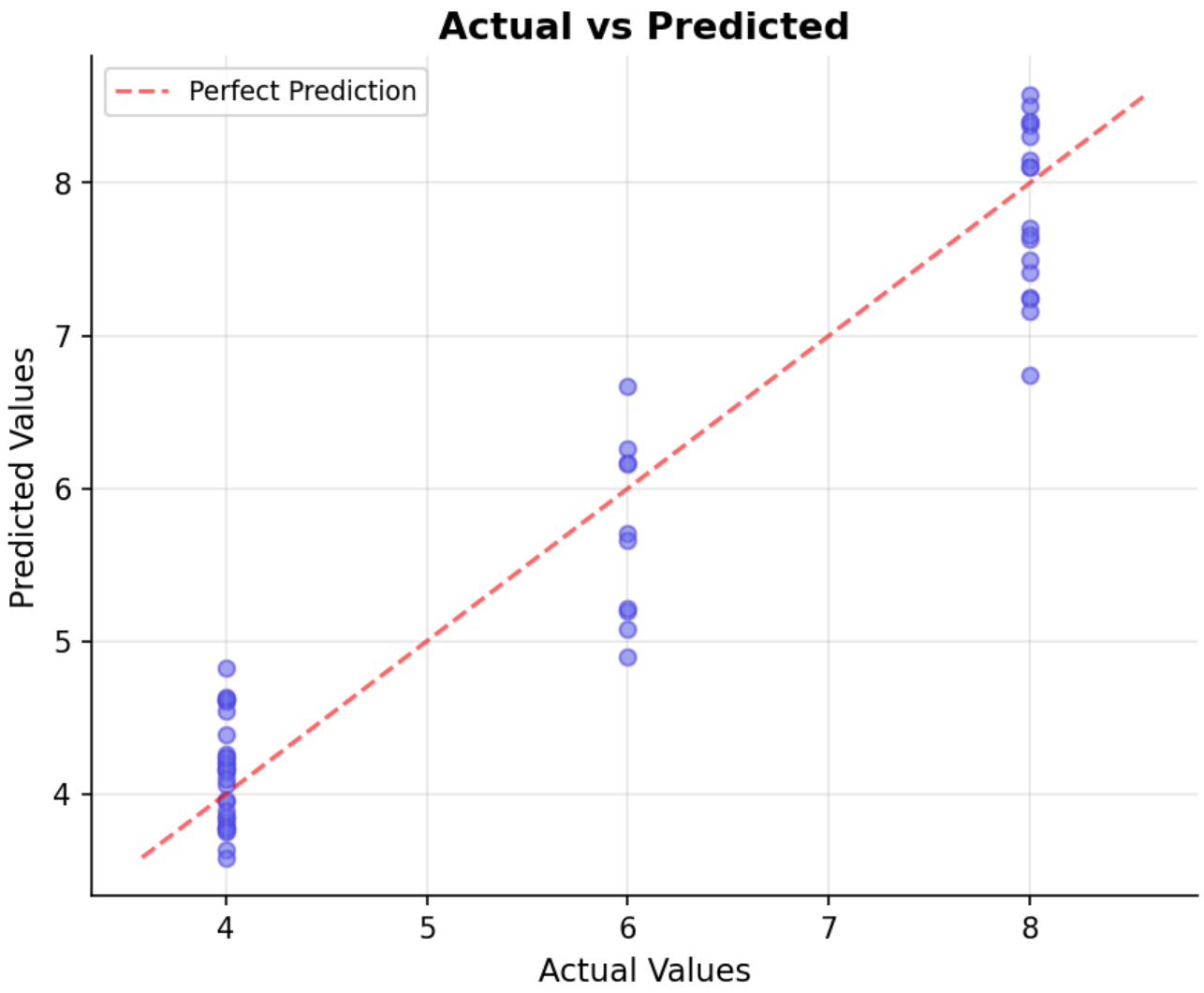
Training Records	**338**
Testing Records	**60**
Prediction Date	**August 02, 2026 at 03:33 PM**

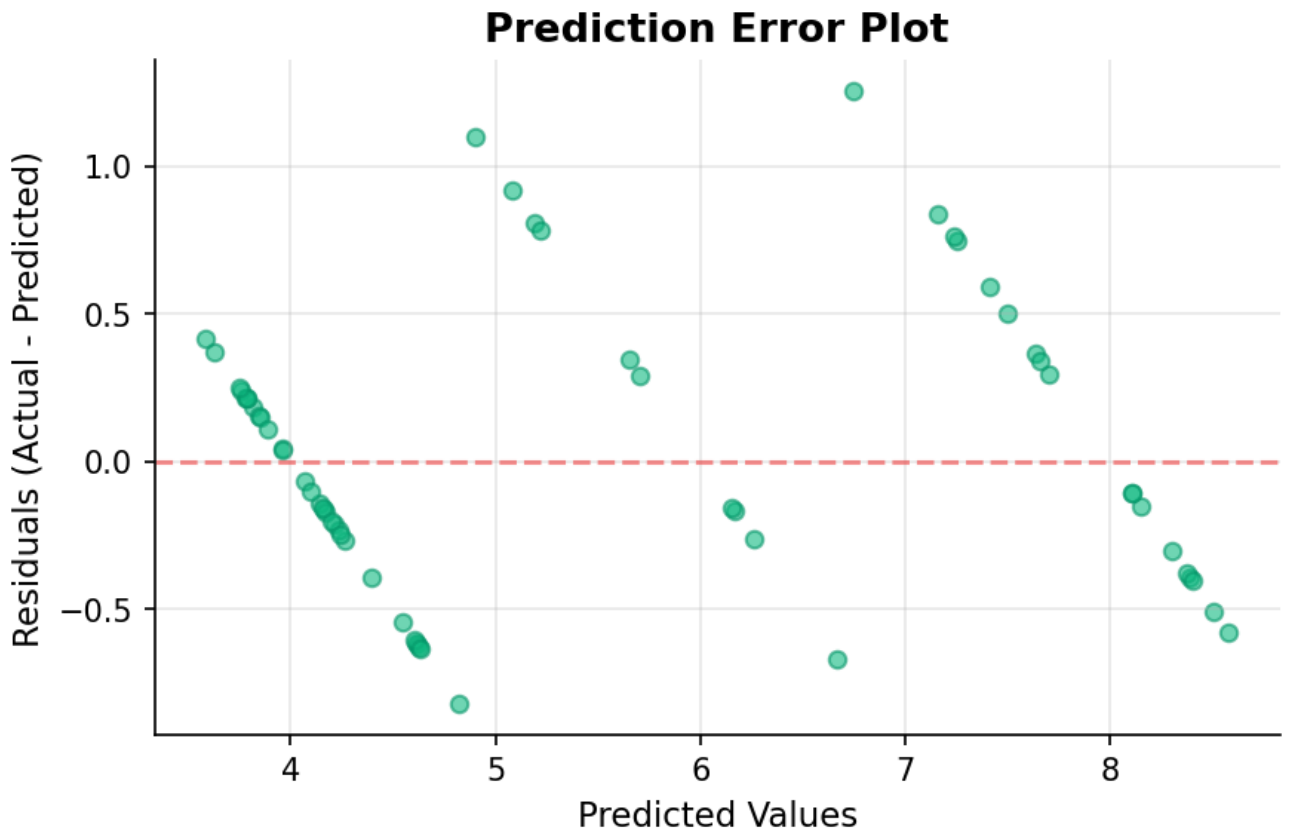
Evaluation Metrics

Regression Metrics

MAE (Mean Absolute Error)	**0.39**
MSE (Mean Squared Error)	**0.23**
R2 Score	**0.9265**

Prediction Visualizations


Actual vs Predicted
Perfect Prediction
Predicted Values
Actual Values
4
5
6
7
8



Prediction Sample

#	Actual Value	Predicted Value
1	4.0	3.8131
2	4.0	4.2654
3	6.0	6.2627
4	8.0	7.7029
5	8.0	7.6339
6	4.0	3.7835
7	4.0	4.613
8	8.0	8.5806
9	8.0	6.7438
10	6.0	5.7079
11	8.0	8.1512

12	4.0	3.9569
13	4.0	3.6308
14	8.0	7.4101
15	4.0	4.6178
16	8.0	8.3923
17	4.0	4.1635
18	6.0	5.0811
19	8.0	8.1046
20	4.0	3.7587

Showing 20 of 60 prediction samples

Model Status

Training Completed Successfully

Generated on August 02, 2026 at 09:04:07 PM